ELEC5206 Laboratory Report
Lab 3 PV Simulation

Bench	4	Date	11 Sep 2026
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1. Introduction

This lab develops a simulation model for the BP SX-10 photovoltaic panel using the ideal single-diode model. The model is used to reproduce the electrical characteristics of the PV panel and to investigate how solar irradiance and cell temperature affect its current, voltage and power output. MATLAB and Simulink are used for parameter calculation, model implementation and simulation.

2. PV Cell Parameterisation

The BP SX-10 panel consists of 36 identical PV cells connected in series. At standard test conditions, the panel parameters are:

Table 1. BP SX-10 panel and equivalent cell parameters at STC.

I_{MS}	V_{MS}	I_{MSC}	V_{MOC}	$Cells$
0.59 A	16.8 V	0.65 A	21.0 V	36
I_{CMS}	V_{CMS}	I_{CSC}	V_{COC}	***Cell***
0.59	0.4667	0.65	0.5833	1

$$I_{CMS} = I_{MS} = 0.59\ A \qquad V_{CMS} = \frac{16.8}{36} = 0.4667\ V$$

$$I_{CSC} = I_{MSC} = 0.65\ A \qquad V_{COC} = \frac{21.0}{36} = 0.5833\ V$$

3. Model Parameter Identification

The ideal single-diode model for one PV cell is

$$i_{pv} = i_{ph} - i_s\left[exp\left(\frac{qv_{pv}}{kT_C A_n}\right) - 1\right]$$

At standard test conditions,

$q = 1.6 \times 10^{-19}$ C ; $k = 1.38 \times 10^{-23}$ J/K ;

$T_{CS} = 298$ K ; $E_{STC} = 1000$ W/m^2;

The two unknown parameters are the diode ideality factor (A_n) and the diode saturation current (i_s). They can be obtained from the open-circuit and maximum-power-point conditions:

$$I_{CSC} = i_s\left[exp\left(\frac{V_{COC}}{V_{TCS}A_n}\right) - 1\right]$$

$$I_{CMS} = I_{CSC} - i_s\left[exp\left(\frac{V_{CMS}}{V_{TCS}A_n}\right) - 1\right]$$

$$V_{TCS} = \frac{kT_{CS}}{q} \approx 0.0257\ V$$

$$A_n \approx 1.905, \quad i_s \approx 4.3576 \times 10^{-6}\ A$$

A compact MATLAB implementation is:

```
q = 1.6e-19; k = 1.38e-23; TCS = 298;
VTCS = k*TCS/q; ICSC = 0.65; ICMS = 0.59;
VCOC = 21/36; VCMS = 16.8/36;
fun = @(x) [ICSC - x(2)*(exp(VCOC/(VTCS*x(1)))-1); ...
            ICMS - ICSC + x(2)*(exp(VCMS/(VTCS*x(1)))-1)];
x = fsolve(fun,[1.9,4.4e-6]);
An = x(1); is = x(2);
```

4. PV Characteristics at STC

The BP SX-10 panel was simulated under standard test conditions using the ideal single-diode model. The panel voltage was swept from 0 V to the open-circuit voltage, and the corresponding PV current was calculated. The output power was then obtained from $P_{pv} = V_{pv} I_{pv}$.

The simulated maximum power point was $V_{MPP} = 16.846\ V$, $I_{MPP} = 0.588\ A$, and $P_{MPP} = 9.912\ W$, which closely matched the rated BP SX-10 values.

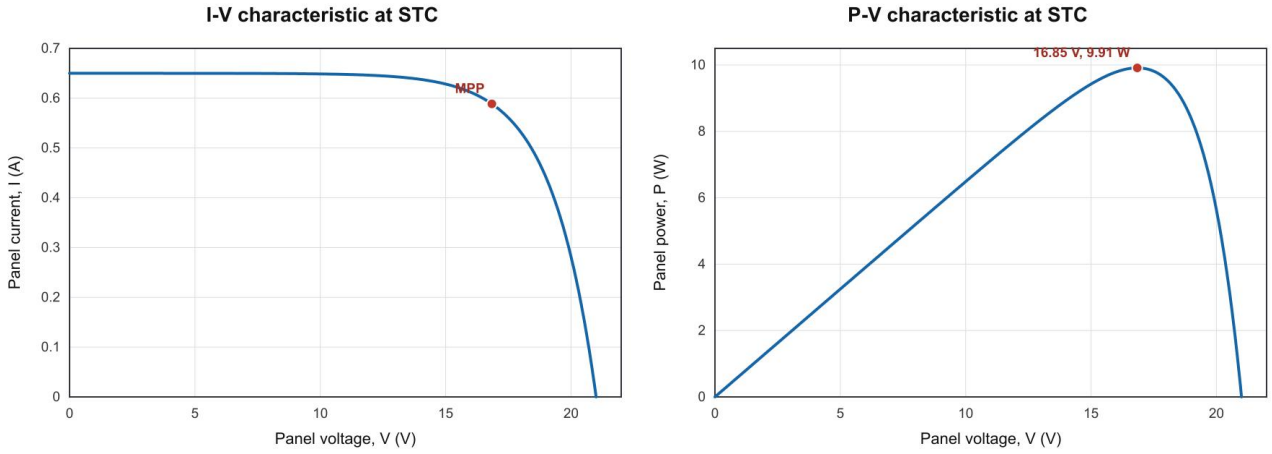


Figure 1. Simulated I-V and P-V characteristics of the BP SX-10 at STC.

5. Simulink Model and Model Evaluation

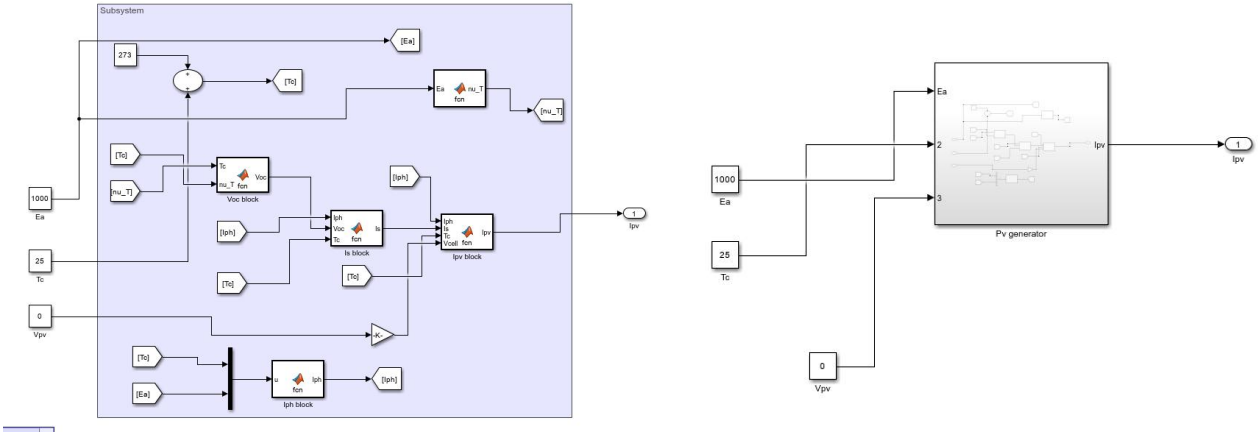


Figure 2. Simulink model of the BP SX-10 PV generator.

The BP SX-10 PV generator was implemented in Simulink using the developed PV cell model. The model takes solar irradiance E_a, cell temperature T_C, and panel voltage V_{pv} as inputs, and calculates the corresponding PV output current I_{pv}.

Table 2. Short-circuit current versus solar irradiance at 25 °C.

Solar irradiance E_a	200 W/m^2	600 W/m^2	800 W/m^2	1000 W/m^2
Current i_{pv} (A)	0.130	0.390	0.520	0.650

The short-circuit current increased almost proportionally with solar irradiance.

Table 3. Short-circuit current versus cell temperature at 1000 W/m².

Temperature T_c	**0 °C**	**25 °C**	**50 °C**
Current i_{pv} (A)	0.639	0.650	0.661

The short-circuit current increased slightly as the cell temperature increased.

Table 4. Output current versus solar irradiance at 25 °C and 18.8 V.

Solar irradiance E_a	200 W/m^2	600 W/m^2	800 W/m^2	1000 W/m^2
Current i_{pv} (A)	0.007	0.187	0.319	0.463

At a panel voltage of 18.8 V, output current increased with irradiance, but less linearly than under short-circuit conditions.

6. Effects of Environmental Conditions

6.1 Effect of Solar Irradiance

With cell temperature held at 25 °C, solar irradiance was varied from 200 to 1000 W/m². The I-V curves show that current increases strongly with irradiance, whereas open-circuit voltage changes only slightly.

The corresponding P-V curves show that maximum output power increases with solar irradiance.

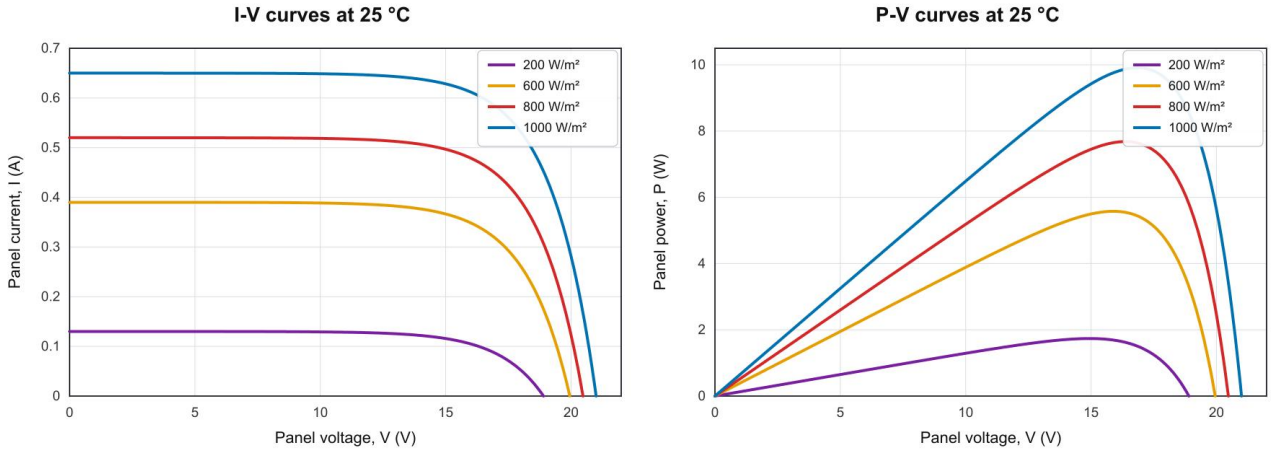


Figure 3. I-V and P-V curves at different solar irradiance levels.

6.2 Effect of Cell Temperature

With solar irradiance held at 1000 W/m², cell temperature was varied from 0 to 50 °C. The I-V curves show a small increase in short-circuit current and a clear reduction in open-circuit voltage as temperature rises.

The P-V curves show that the maximum output power decreases as cell temperature increases.

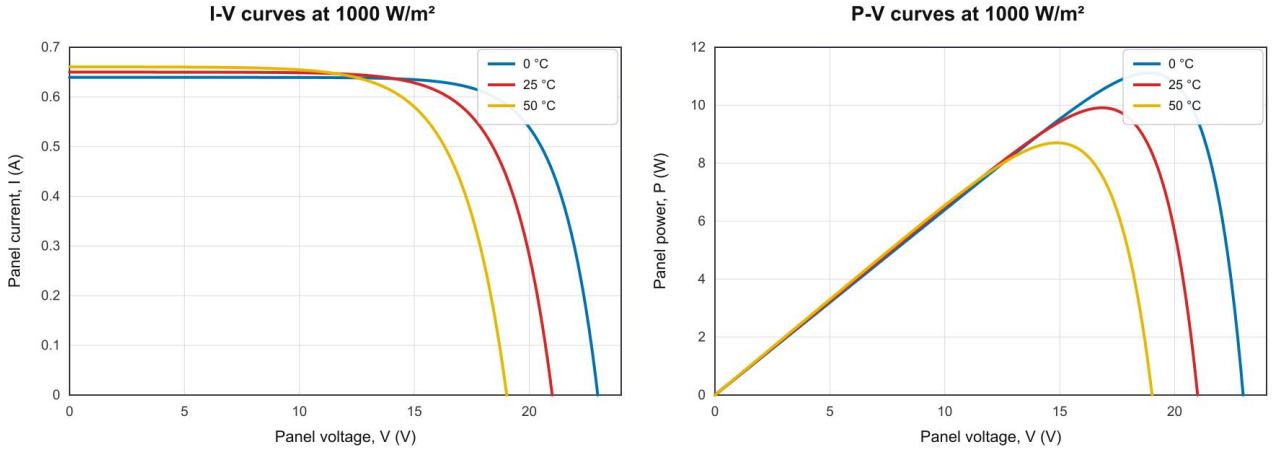


Figure 4. I-V and P-V curves at different cell temperatures.

7. Discussion and Conclusion

7.1 What We Learned

The simulation demonstrated the main operating trends of a photovoltaic module. Irradiance primarily changed the photocurrent and maximum power. Temperature had a smaller effect on short-circuit current, but it shifted the open-circuit

voltage and the position of the maximum power point. These trends are consistent with the single-diode model used in the study.

7.2 Suggestions for Improvement

The ideal model neglects series resistance, shunt leakage and cell mismatch, so differences from a physical panel are most likely near the knee and open-circuit regions. A stronger follow-up study would include these losses and compare the simulated curves with datasheet or measured data under the same irradiance and temperature.

7.3 Conclusion

A MATLAB and Simulink model of the 36-cell BP SX-10 module was parameterised and evaluated. At STC, the simulated maximum power point was approximately 16.846 V, 0.588 A and 9.912 W, closely matching the rated operating point. The environmental sweeps also reproduced the expected increase in current and power with irradiance and the reduction in voltage and maximum power with increasing cell temperature.